Tut-5 Solution (Combination)

Ans 1: (I)

(A) $2^7 \times 2^{10} \times 2^{12} = 2^{29}$ (B) $(2^{29} - 1)$

(C) (B) $(2^7-1) \times (2^{10}-1) \times (2^{12}-1)$ (D) ${}^7C_0 \times 2^{22}$

(E) $({}^7C_1 + {}^7C_2 + \cdots {}^7C_7) = 2^7 - 1$ (F) ${}^7C_7 \times 2^{22}$

(II) (A) ${}^7C_1 \times {}^{10}C_1 \times {}^{12}C_1$ (B) ${}^7C_3 + {}^{10}C_3 + {}^{12}C_3$

(C) Same as (A) ${}^7C_1 \times {}^{10}C_1 \times {}^{12}C_1$

(D) ${}^7C_1 \times {}^{22}C_2 + {}^7C_2 \times {}^{22}C_1 + {}^7C_3 \times {}^{22}C_0$

$= {}^{29}C_3 - {}^7C_0$

(E) ${}^{29}C_3 - [{}^7C_3 + {}^{10}C_3 + {}^{12}C_3]$ (F) ${}^7C_0 \times {}^{22}C_3$

(2) $a_1 a_2 a_3 \cdots a_n$

$a_1, a_2, \cdots a_n$ are strings of length 1

$a_1a_2, a_2a_3, \cdots a_{n-1}a_n$ are substrings of length 2

length of substrings	Number of substrings
1	$\to n$
2	$\to n-1$
3	$\to n-2$
⋮	
n	$\to 1$

Total $= 1 + 2 + \cdots n = \boxed{\frac{n(n+1)}{2}}$

3. n zeros would make $(n+1)$ place next to there. Where we have to put k 1's.

$$= \boxed{{}^{n+1}C_k}$$

4. Distribute 10 Rose × Distribute 15 Sunflower × 14 daffodils

$$= \begin{array}{ll} n_1 = 10, \; r = 2 & \text{Roses.} \\ n_2 = 15, \; r = 2 & \text{Sunflower} \\ n_3 = 14; \; r = 2 & \text{— Daffodils} \end{array}$$

$$= {}^{10+2-1}C_{2-1} \times {}^{15+2-1}C_{2-1} \times {}^{14+2-1}C_{2-1}$$

5. Each couple

		Husband	wife
	Not allowed ←	Comes	don't come
1 ←	Allowed ←	don't come	comes
2 ←	Allowed ←	comes	comes
3 ←	Allowed ←	don't come	don't come

Thus each couple has 3 options

Total n couples have $3 \times 3 \cdots n$
$= 3^n$

6. $n^* = m - kn$
$r = n$

$$= {}^{(m-kn)+n-1}C_{n-1}$$

7 Deaarrangement of 5 things

$$D_5 = 5!\left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right]$$

$$= 5!\left[\frac{60 - 20 + 5 - 1}{5!}\right] = 44$$

8 To Reach (0,0) to (10,10) only Right or Up steps allowed.

Total 10 Rightsteps and 10 up steps.

Thus I. $\boxed{{}^{20}C_{10}}$

II. We don't need to take (4,5) to (5,4) Thus

To Reach (0,0) to (4,4) to (5,4) to (10,10)

$= {}^{8}C_4 \times 1 \times {}^{11}C_5$

$\Rightarrow$ Thus Total - to Reach from (4,5) to (5,4)

$$= \boxed{{}^{20}C_{10} - {}^{8}C_4 \times 1 \times {}^{11}C_5}$$

9 $\boxed{{}^{b+n-1}C_{n-1} \times {}^{r+n-1}C_{n-1}}$

10 (a) 10 → 1's

(b) 8 → 1's, 1-2's

(c) 6 → 1's, 2-2's

(D) 4 → 1's, 3-2's

(E) 2-1's, 4-2's

(F) 5 → 2's

Total ~~to~~ ~~${}^{10}C_2 + {}^{9}C_2 + {}^{8}C$~~

Then total

$$^{10}C_0 + {}^{9}C_1 + {}^{8}C_2 + {}^{7}C_3 + {}^{6}C_4 + {}^{5}C_5$$

$$\Rightarrow 1+9+28+35+15+1 = 89$$

Ans (11) 50th Toss is 20th Head

Then in FIRST 49 tosses, Exactly 19 Heads

and In last 50 Tosses Exactly 10 Heads.

$$= {}^{49}C_{19} \times {}^{50}C_{10}$$

(12) $\dfrac{40 \times 36}{2}$ $\dfrac{\text{FIRST PERSON Then Second}}{2 \text{ (ORDER DOES NOT matter)}}$

(13) Total line $= {}^{20}C_2$

line due to collinear $= {}^{7}C_2$ (Remove)

$\Rightarrow {}^{20}C_2 - {}^{7}C_2 + 1 \rightarrow$ one line due to all collinear point

(14) $n^* = 0 - (-10 - 7 + 5) = 12$

$r = 3$

Then ${}^{12+3-1}C_{3-1} \Rightarrow {}^{14}C_2$

(15) $\boxed{{}^{5}C_2 \times \lfloor 4}$

which two assignment given to same person.

(16) Case 1: Teams played only 3 game.

AAA, BBB → 2

Case 2: Teams played exactly 4 game

2A, 1B — — — A OR — — 2B, 1A — B

$^3C_1 + {}^3C_1 = 6$

Case 3: Team played Exactly 5 games

2A, 2B — — — A OR B ⇒ $^4C_2 \times 2 \Rightarrow 12$

(20) = 2+6+12

(17) $^{10}C_2 - 10$

(18) $\lfloor 5-1 \times \lfloor 5$

(19) (I) Select 4 & 5 persons $^9C_4$ or $^9C_5$

Arrange 5 persons on one side

4 person on another side

$^9C_4 \times {}^5P_5 \times {}^5P_4 =$

(II) Case 1: All girls inside when 4 person

$\lfloor 5 \lfloor 5$

Case 2: 1B & 4G on one side

$^5C_4 \times {}^5P_4 \times {}^5P_5$

$\Rightarrow \lfloor 5 \lfloor 5 + {}^5C_4 \times \lfloor 5 \lfloor 5 =$

(20) Tie two Boys. so Exactly 5 groups.

$= \lfloor 5-1 \times \lfloor 2$.

$= \lfloor 4 \lfloor 2$.